

BdOI Monthly Contest Round1 Solutions:

Problem Blackout Math

Problem Information

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Subtasks hinting key observations

Preliminary Observation:

The sequence is defined as

$$F_1 = A$$

$$F_2 = B$$

$$F_i = F_{i-1} \oplus F_{i-2} \quad \text{for all } i \geq 3$$

Since N can be as large as 10^{18} , we clearly cannot simulate the recurrence term by term for the full solution within the time constraints. We need to find a efficient solution to get a full score. However a basic simution will award you the partial points.

Subtask 1: $N \leq 3$

This is a very easy subtask. For $N \leq 2$ we just need need print the first and second number in the sequence depending on the input of N . For $N = 3$ we just need to print $A \oplus B$.

Subtask 2: $A = 1$ and $B = 1$ and $N \leq 10$

In this subtask , we can simulate N^{th} term in the problem by writing a recursive function.

Subtask 3: $N \leq 20$

same as subtask 2.

Subtask 4: $N \leq 1000$

In this subtask , we can simulate N^{th} term in the problem by writing a recursive function. However, it is not efficient always. It can cause stack overflow and give you error. A simple approach to use a loop to simulate the sequence.

Subtask 5: $N \leq 10^5$

same as subtask 4.

Subtask 6(Full solution)

Here N is far too large to simulate directly. If we do some dry runs we will find a pattern. The key property to notice here the functionality and property of the XOR($\oplus$) operator. Two of its important properties we need to notice are $x \oplus x = 0$ and $x \oplus 0 = x$. Let's use this to compute a few terms and see what happens:

$$\begin{aligned}F_3 &= F_2 \oplus F_1 = A \oplus B \\F_4 &= F_3 \oplus F_2 = (A \oplus B) \oplus B = A \oplus (B \oplus B) = A \\F_5 &= F_4 \oplus F_3 = A \oplus (A \oplus B) = (A \oplus A) \oplus B = B \\F_6 &= F_5 \oplus F_4 = B \oplus A = A \oplus B \\F_7 &= F_6 \oplus F_5 = (A \oplus B) \oplus B = A \oplus (B \oplus B) = A\end{aligned}$$

Turns out the There is a repeated sequence throughout the terms, we can take this advantage to Optimize our solution.

This pattern repeats in every 3 sequence!

The sequence is periodic with period 3, cycling through $A, B, A \oplus B, A, B, A \oplus B, \dots$ forever.

With this simple observation our problem simplifies to the idea below:

$$F_N = \begin{cases} A & \text{if } N \bmod 3 = 1 \\ B & \text{if } N \bmod 3 = 2 \\ A \oplus B & \text{if } N \bmod 3 = 0 \end{cases}$$

This reduces the time complexity and gives a solution that score full points.